

● EASY

● ALGEBRA

Linear functions

08

8 questions · ~12 min

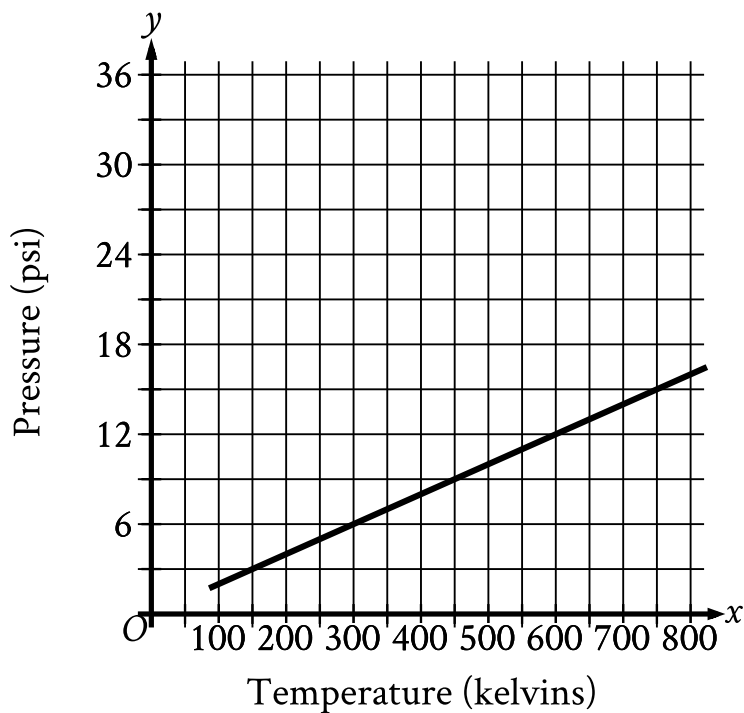
Q1

ALGEBRA

The length, y , of a white whale was 162 centimeters cm when it was born and increased an average of 4.8 cm per month for the first 12 months after it was born. Which equation best represents this situation, where x is the number of months after the whale was born and y is the length, in cm, of the whale?

- A. $y = 162x$
- B. $y = 162x + 162$
- C. $y = 4.8x + 4.8$
- D. $y = 4.8x + 162$

Argon is placed inside a container with a constant volume. The graph shows the estimated pressure y , in pounds per square inch psi, of the argon when its temperature is x kelvins.



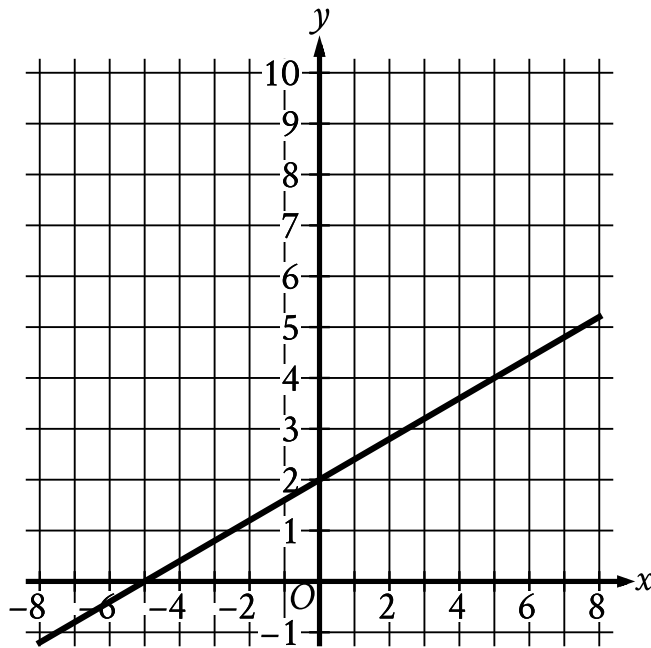
- The line slants gradually up from left to right.
- The line begins at the approximate point (90 comma 1.8).
- The line passes through the following points:
 - (300 comma 6)
 - (450 comma 9)
 - (600 comma 12)

What is the estimated pressure of the argon, in psi, when the temperature is 600 kelvins?

- A. 6
- B. 12

C. 300

D. 600



- The line slants gradually up from left to right.
- The line passes through the following points:
 - $(-5, 0)$
 - $(0, 2)$

The graph of the linear function f is shown. What is the y -intercept of the graph of $y = fx$?

- A. $-5, 0$
- B. $2, 0$
- C. $0, 2$
- D. $0, -5$

Q4

ALGEBRA

On January 1, 2015, a city's minimum hourly wage was \$9.25. It will increase by \$0.50 on the first day of the year for the next 5 years. Which of the following functions best models the minimum hourly wage, in dollars, x years after January 1, 2015, where $x = 1, 2, 3, 4, 5$?

- A. $f(x) = 9.25 - 0.50x$
- B. $f(x) = 9.25x - 0.50$
- C. $f(x) = 9.25 + 0.50x$
- D. $f(x) = 9.25x + 0.50$

Q5

ALGEBRA

The function f is defined by $fx = \frac{1}{2}x + 6$. What is the value of $f4$?

- A. 20
- B. 12
- C. 10
- D. 5

Q6

ALGEBRA

The function h is defined by $hx = 3x - 7$. What is the value of $h-2$?

- A. -13
- B. -10
- C. 10
- D. 13

Q7

ALGEBRA

To repair a refrigerator, a technician charges \$ 60 per hour for labor plus \$ 120 for parts. Which function f represents the total amount, in dollars, the technician will charge for this job if it takes x hours?

- A. $fx = x + 120$
- B. $fx = 60x$
- C. $fx = 60x + 120$
- D. $fx = 60x - 120$

For the linear function f , $f0 = 17$ and $f1 = 17$. Which equation defines f ?

A. $fx = \frac{1}{17}$

B. $fx = 1$

C. $fx = 17$

D. $fx = 34$